

Lyna Hocine

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Education

Sorbonne University <i>M2 Algorithms, Artificial Intelligence, Interactions and Decision (AI2D)</i> • M1 ranking: 10/43 • Relevant Coursework: Reinforcement Learning & Robotics, Human-Machine Interaction, Foundations of Multi-Agent Systems, Statistical & Probabilistic Models, Multi-Agent Coordination & Consensus	Sep 2025 – Present Paris, France
University of Milan – 4EU+ Erasmus+ BIP <i>From Data to Systems: Designing and Deploying AI in Real-World Contexts</i> • Intensive training on designing and deploying AI systems in real-world contexts, with a focus on challenges arising from real-world data.	Jun 2026 Milan, Italy
Johannes Kepler University <i>Exchange Semester – Computer Science</i> • Relevant Coursework: Unsupervised Machine Learning, SAT Solving, Engineering of AI-Intensive Systems	Feb 2025 – Jun 2025 Linz, Austria
Sorbonne University <i>Bachelor's Degree in Computer Science</i>	Sep 2021 – Jun 2024 Paris, France

Research Experience

ISIR – Sorbonne University / CNRS <i>Research Intern – Human-Swarm Interaction</i> • Investigated human-swarm interaction through intuitive hand gestures for controlling collective robot behaviors. • Extended a C++ swarm robotics simulator, working with its runtime components for robot dynamics, proximity sensing, and inter-robot communication. • Implemented and evaluated decentralized collective behaviors including heading alignment, target tracking, formation control, and leader-follower coordination. • Investigated the limitations of local proximity sensing for interpreting directional gestures under noisy and incomplete state information.	Jun 2026 – Jul 2026 Paris, France
ISIR – Sorbonne University <i>Research Project – HARMONI + MCP</i> • Integrated HARMONI, a multimodal personalization framework for multi-user human-robot interaction, with the Model Context Protocol (MCP) to expose its capabilities as structured tools for LLM-based agents. • Designed a context-aware interaction pipeline connecting LLM reasoning with multimodal robot context, enabling memory-informed interactions on Enchanted Tools' Mirokaï robotic platform. • Implemented the MCP middleware in Python using FastMCP, LangChain, and Ollama for local LLM-based tool use.	2026 Paris, France
Sorbonne University <i>Research Project – LLM-Driven Reward Function Generation for Reinforcement Learning</i> • Conducted a literature-based study of LLM-generated reward functions for reinforcement learning, focusing on the EUREKA framework. • Analyzed prompt synthesis, automated feedback mechanisms, evolutionary search, and the interaction between generated rewards and policy optimization. • Investigated the potential of LLMs to reduce the manual reward-engineering bottleneck in robot learning.	2026 Paris, France

Technical Skills

Programming: Python, C++, C, Java, OCaml, SQL
ML / AI: PyTorch, NumPy, Pandas, scikit-learn
Robotics / Simulation: Swarm robotics, multi-agent simulation
AI Systems: MCP, FastMCP, LangChain, Ollama
Tools: Git, Linux, Bash
Languages: Arabic (Native), French (C1), English (C1), German (A1)

Research Interests

Interactive Robot Learning, Human-Robot Interaction, Reinforcement Learning, Multi-Agent Systems, Embodied AI